

RIDHI BANDARU

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she/they

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Education

University of Connecticut, Storrs, CT

M.S., Data Science, 2024

GPA: 3.93/4.00

Birla Institute of Technology and Science, Pilani – Hyderabad Campus, Hyderabad, India

B.E., Electronics and Instrumentation Engineering (Honors), 2023

Minor, Robotics and Automation

CGPA: 8.27/10.00

Research Experience

Language of Thought Modeling for Sequence Generation

February 2025 – Present

w/ Dr. Abhishek Dedhe & Prof. Jessica Cantlon

Kid Neuro Lab, Carnegie Mellon University, Pittsburgh

- Designed and implemented a Language-of-Thought framework with a cognitively grounded Domain-Specific Language to model hierarchical sequence-generation strategies
- Developed methods to estimate algorithmic/cognitive complexity of generative strategies using Minimum Description Length, K-Complexity, and related metrics
- Led a full program-synthesis approach, enabling systematic exploration of the entire hypothesis space
- Preparing two papers: a short conference submission and a full-length journal manuscript

Sequence Generation Biases across Language / Vision-Language Models

October 2025 – Present

w/ Mukund Choudhary & Prof. Monojit Choudhury

MBZUAI, Abu Dhabi

- Extending behavioral sequence-processing tasks to large language models and vision-language models
- Translate experimental paradigms into model-evaluation pipelines
- Submitted LLM component research (under review)
- Designing, running, and analyzing all VLM experiments

Papers

Bandaru, R., Dedhe, A.M., Lee, B., Xiong, M., Ferrigno, S.M., Piantadosi, S.T., & Cantlon, J.F. (*In prep*). **One Cognitively Plausible Language of Thought for Hierarchical Sequence Generation.**

Choudhary, M., **Bandaru, R.,** Agrawal, A., Dedhe, A.M., Cantlon, J.F., & Choudhury, M. (*In prep*). **Parallels in Hierarchical Composition: Human and Machine Sequence Processing Compared.**

Choudhary, M., Agrawal, A.*, **Bandaru, R.*¹**, Choudhury, M., & Dedhe, A.M. (*Under review*). **Stochastic Parrot See, Stochastic Parrot Do: Hierarchical Sequence Processing Across Artificial and Biological Intelligences.**

Bandaru, R., & Dedhe, A.M. (*Under review*). **The price of hierarchical reasoning** [Extended abstract].

Talks / Presentations

Bandaru, R. LoT-Modeling Sequential Generation Tasks.

October 1, 2025. Presented to Dr. Steve Piantadosi, PI, Computation and Language Lab, University of California, Berkeley

October 23, 2025. Talk at the Concepts, Actions, and Objects Lab, Carnegie Mellon University.

¹ * = equal contribution. Order determined alphabetically by last name

Awards / Scholarships

Independent Study Grant of \$1325 (Fall 2023), Department of Chemistry, University of Connecticut

Annual Departmental Scholarship of \$7855 (Fall 2023), Department of Applied Mathematics and Statistics, Johns Hopkins University (did not accept)

Undergraduate Teaching Award of ₹20000 (Spring 2023), Academic-Undergraduate Studies Division, BITS Pilani – Hyderabad Campus

Teaching / Mentorship

Mentor (Summer 2025, Fall 2025), **Kid Neuro Lab, Carnegie Mellon University / Troy High School CA**

Students: Helen Fang, Brandon Lee, Michael Xiong

- Mentored students from a nationally ranked magnet school during their research internship
- Led reading discussions on cognitive modeling methodology for behavioral analysis and LoT
- Supervised programming tasks for DSL complexity analyses and module design for the existing codebase

Tutor (Spring 2024), **Undergraduate Math & Statistics, Q-Center, University of Connecticut**

- Provided one-on-one and group tutoring sessions to undergraduate students in graduate-level courses
- Developed weekly mini-lessons and offered structured support for assignments and exam preparation

Teaching Assistant (Spring 2023), **CS F327 Computer Architecture, BITS Pilani, Hyderabad Campus**

Lab, 40 students

- Assisted students with Verilog simulations of pipelined processors

Languages

Programming: Python (Torch), R, MATLAB, C, C++, HTML/CSS, JavaScript, LaTeX

Natural Languages: English, Hindi-Urdu, Telugu, French (Functional), Latin (Basic)

Writing systems: Latin, Devanagari, Telugu, Perso-Arabic, Brahmi

Industry Experience

Software Engineer

Centene Corporation

September 2024 – Present

St. Louis, MO

- Architecting scalable graph database solutions (with Neo4j) to model high-dimensional healthcare data

Data Scientist

Indeed, Inc.

May 2024 – July 2024

Stamford, CT

- Developed predictive models to analyze job postings, applications, market, and user behavior, utilizing feature engineering techniques to improve model accuracy and interpretability

Software Engineering Intern

Dell EMC

July 2022 – December 2022

Bengaluru, India

- Engineered a decision tree-based classification system for real-time device health monitoring to predict hardware failures proactively

Summer Intern

Edutech Learning Solutions

May 2021 – July 2021

Vadodara, India

- Developed a computer vision-based people counting system, using object detection algorithms to accurately track and count individuals entering and leaving a space in real-time

Service / Outreach

Lambada Physical Reasoning Dataset Construction, 2025

- Designed and built a culturally grounded PIQA²-style physical reasoning dataset for an extremely low-resource language, enabling evaluation of commonsense reasoning in underrepresented linguistic settings.

² Bisk, Y., Zellers, R., Bras, R. L., Gao, J., & Choi, Y. (2019). PIQA: Reasoning about Physical Commonsense in Natural Language (Version 1). arXiv. <https://doi.org/10.48550/ARXIV.1911.11641>

- Recruited and coordinated a team of 10 native speakers to translate and validate prompts, creating one of the first structured reasoning datasets available for this language.
- Contributed to expanding the representation of marginalized languages in commercial AI systems by delivering high-quality annotations and a standardized dataset pipeline

Panini Linguistics Olympiad (PLO)

Problemsetter, 2025 & 2024

Student-Faculty Council, BITS Pilani – Hyderabad Campus

Acted as a mediator between departmental faculty and over 500 students for a selection of courses, to discuss ways to make the student experience better during the remote mode of study during the COVID-19 era

HSS F222 Introduction to Linguistics (Spring 2023)

CS F327 Computer Architecture (Spring 2022)

EEE F211 Electrical Machines (Spring 2021)

Rotaract Club, BITS Pilani – Hyderabad Campus, 2021

Hosted a professional development workshop for girls in STEM, with Chair, Grace Hopper Conference India

References

Prof. Jessica Cantlon

Ronald J. and Mary Ann Zdrojowski Professor of Developmental Neuroscience
Carnegie Mellon University
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Dr. Abhishek Dedhe

Postdoc, Wise Judgement Consortium
University of Waterloo
adedhe@uwaterloo.ca